

Statistics and Probability (10 th std)

Choose the Appropriate Answer

1. Which of the following is not a measure of dispersion?

- (a) Range (b) Standard deviation
(c) Arithmetic mean (d) Variance

Ans: (c) Arithmetic mean

2. The range of the data 8, 8, 8, 8, 8... 8 is

- (a) 0 (b) 1 (c) 8 (d) 3

Ans: (a) 0

3. The sum of all deviations of the data from its mean is _____

- (a) Always positive (b) Always negative
(c) Zero (d) Non - zero integer

Ans: (c) Zero

4. The mean of 100 observations is 40 and their standard deviation is 3. The sum of squares of all deviations is

- (a) 40000 (b) 160900 (c) 160000 (d) 30000

Ans: (b) 160900

$$\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 = 9 \Rightarrow \frac{\sum x^2}{100} - 40^2 = 9$$

$$\frac{\sum x^2}{100} = 9 + 1600 \Rightarrow \frac{\sum x^2}{100} = 1609 \Rightarrow \sum x^2 = 160900$$

5. Variance of first 20 natural numbers is _____

- (a) 32.25 (b) 44.25 (c) 33.25 (d) 30

Ans: (c) 33.25

$$\text{Variance of 20 natural numbers is} = \frac{20^2 - 1}{12} = \frac{400 - 1}{12} = \frac{399}{12} = 33.25$$

6. The standard deviation of a data is 3. If each value is multiplied by 5 then the new variance is

- (a) 3 (b) 15 (c) 5 (d) 225

Ans: (d) 225

$$\text{The new S.D} = 5 \times 3 = 15 \quad \text{Variance} = 15^2 = 225$$

7. If the standard deviation of x, y, z is p then the standard deviation of 3x + 5, 3y + 5, 3z + 5 is _____

- (a) 3p + 5 (b) 3p (c) p + 5 (d) 9p + 15

Ans: (b) 3p

8.If the mean and coefficient of variation of a data are 4 and 87.5% then the standard deviation is

- (a) 3.5 (b) 3 (c) 4.5 (d) 2.5

Ans: (a) 3.5

$$\begin{aligned} \text{C.V} &= \frac{\sigma}{\bar{x}} \times 100 \Rightarrow 87.5 = \frac{\sigma}{4} \times 100 \\ \sigma &= \frac{87.5 \times 4}{100} \Rightarrow = 3.5 \end{aligned}$$

9.Which of the following is incorrect?

- (a) $P(A) > 1$ (b) $0 \leq P(A) \leq 1$ (c) $P(\Phi) = 0$ (d) $P(A) + P(\bar{A})=1$

Ans: (a) $P(A) > 1$

10.The probability a red marble selected at random from a jar containing p red, q blue and r green marbles is _____

- (a) $q / (p + q + r)$ (b) $p / (p + q + r)$
(c) $(p + q) / (p + q + r)$ (d) $(p + r) / (p + q + r)$

Ans: (a) $q / (p + q + r)$

11.A page is selected at random from a book. The probability that the digit at units place of the page number chosen is less than 7 is _____

- (a) $3 / 10$ (b) $7 / 10$ (c) $3 / 9$ (d) $7 / 9$

Ans: (b) $7 / 10$

12.The probability of not getting the job is then the value of x is.

- (a) 2 (b) 1 (c) 3 (d) 1.5

Ans: (b) 1

13.Kamalam went to play a lucky draw contest. 135 tickets of the lucky draw were sold. If the probability of Kamalam winning is $1 / 9$, then the number of tickets bought by Kamalam is _____

- (a) 5 (b) 10 (c) 15 (d) 20

Ans: (c) 15

14.If a letter is chosen at random from the English alphabets {a, b, ..., z}, then the probability that the letter chosen precedes x.

- (a) $12 / 13$ (b) $1 / 13$ (c) $23 / 26$ (d) $3 / 26$

Ans: (c) $23 / 26$

15.A purse contains 10 notes of ₹ 2000, 15 notes of ₹ 500, and 25 notes of ₹ 200. One note is drawn at random. What is the probability that the note is either a ₹ 500 note or ₹ 200 note?

- (a) $1 / 5$ (b) $3 / 10$ (c) $2 / 3$ (d) $4 / 5$

Ans: (d) $4 / 5$

16.The probability of event equal to zero is called;

- (a) Unsure event (b) Sure Event
(c) Impossible event (d) Independent event

Ans: (c) Impossible event

17. The probability that cannot exist among the following:

- (a) $\frac{2}{3}$ (b) -1.5 (c) 15% (d) 0.7

Ans: (b) -1.5

18. If $P(E) = 0.07$, then what is the probability of 'not E'?

- (a) 0.93 (b) 0.95 (c) 0.89 (d) 0.90

Ans: (a) 0.93

19. A bag has 3 red balls and 5 green balls. If we take a ball from the bag, then what is the probability of getting red balls only?

- (a) 3 (b) 8 (c) $\frac{3}{8}$ (d) $\frac{8}{3}$

Ans: (c) $\frac{3}{8}$

20. A bag has 5 white marbles, 8 red marbles and 4 purple marbles. If we take a marble randomly, then what is the probability of not getting purple marble?

- (a) 0.5 (b) 0.66 (c) 0.08 (d) 0.77

Ans: (d) 0.77

21. A dice is thrown in the air. The probability of getting odd numbers is

- (a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) 3 (d) 4

Ans: (a) $\frac{1}{2}$

22. If we throw two coins in the air, then the probability of getting both tails will be:

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) 2 (d) 4

Ans: (b) $\frac{1}{4}$

23. If two dice are thrown in the air, the probability of getting sum as 3 will be

- (a) $\frac{2}{18}$ (b) $\frac{3}{18}$ (c) $\frac{1}{18}$ (d) $\frac{1}{36}$

Ans: (c) $\frac{1}{18}$

24. A card is drawn from the set of 52 cards. Find the probability of getting a queen card.

- (a) $\frac{1}{26}$ (b) $\frac{1}{13}$ (c) $\frac{4}{53}$ (d) $\frac{4}{13}$

Ans: (b) $\frac{1}{13}$

25. A fish tank has 5 male fish and 8 female fish. The probability of fish taken out is a male fish:

- (a) $\frac{5}{8}$ (b) $\frac{5}{13}$ (c) $\frac{13}{5}$ (d) 5

Ans: (b) $\frac{5}{13}$

26. The sum of the probabilities of all the elementary events of an experiment is

- (a) 0.5 (b) 1 (c) 2 (d) 1.5

Ans: (b) 1

27. A card is selected at random from a well shuffled deck of 52 playing cards. The probability of its being a face card is

- (a) $\frac{3}{13}$ (b) $\frac{4}{13}$ (c) $\frac{6}{13}$ (d) $\frac{9}{13}$

Ans: (a) $\frac{3}{13}$

28.If an event cannot occur, then its probability is

- (a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) 0

Ans: (d) 0

29.An event is very unlikely to happen. Its probability is closest to

- (a) 0.0001 (b) 0.001 (c) 0.01 (d) 0.1

Ans: (a) 0.0001

30.If $P(A)$ denotes the probability of an event A, then

- (a) $P(A) < 0$ (b) $P(A) > 1$ (c) $0 \leq P(A) \leq 1$ (d) $-1 \leq P(A) \leq 1$

Ans: (c) $0 \leq P(A) \leq 1$

31.The probability that a non-leap year selected at random will contain 53 Sundays is

- (a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{5}{7}$

Ans: (a) $\frac{1}{7}$

32.If the probability of an event is p , the probability of its complementary event will be

- (a) $p - 1$ (b) p (c) $1 - p$ (d) $1 - \frac{1}{p}$

Ans: (c) $1 - p$

33.A card is drawn from a deck of 52 cards. The event E is that card is not an ace of hearts. The number of outcomes favourable to E is

- (a) 4 (b) 13 (c) 48 (d) 51

Ans: (d) 51

34.The probability of getting a bad egg in a lot of 400 is 0.035. The number of bad eggs in the lot is

- (a) 7 (b) 14 (c) 21 (d) 28

Ans: (b) 14

35.Two players, Sangeeta and Reshma, play a tennis match. It is known that the probability of Sangeeta winning the match is 0.62. The probability of Reshma winning the match is

- (a) 0.62 (b) 0.38 (c) 0.58 (d) 0.42

Ans: (b) 0.38

36.An event is very unlikely to happen. Its probability is closest to:

- (a) 0.0001 (b) 0.001 (c) 0.01 (d) 0.1

Ans: (a) 0.0001

37.If the probability of an event is P , the probability of its complementary event will be:

- (a) $P - 1$ (b) P (c) $1 - p$ (d) $1 - \frac{1}{p}$

Ans: (c) $1 - p$

38.In a family of 3 children, the probability of having at least one boy is:

- (a) $\frac{7}{8}$ (b) $\frac{1}{8}$ (c) $\frac{5}{8}$ (d) $\frac{3}{4}$

Ans: (d) $\frac{3}{4}$

39.If $P(A)$ denotes the probability of an event then:

- (a) $P(A) < 0$ (b) $P(A) > 0$ (c) $0 \leq P(A) \leq 1$ (d) $-1 \leq P(A) \leq 0$

Ans: (c) $0 \leq P(A) \leq 1$

40.The probability than a non-leap year selected at random will contain 53 Sundays is:

- (a) $1/7$ (b) $2/7$ (c) $3/7$ (d) $5/7$

Ans: (a) $1/7$

41.When a die is thrown, the probability of getting an odd number less than 3 is:

- (a) $1/6$ (b) $1/3$ (c) $1/2$ (d) 0

Answer: (a) $1/6$

42.A card is drawn from a deck of 52 cards. The event E is that card is not an ace of hearts. The number of outcomes favourable to E is:

- (a) 4 (b) 13 (c) 48 (d) 51

Ans: (d) 51

43.The probability of getting a bad egg in a lot of 400 is 0.035. The number of bad eggs in the lot is:

- (a) 7 (b) 14 (c) 21 (d) 28

Ans: (b) 14

44.A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought?

- (a) 40 (b) 240 (c) 480 (d) 750

Ans: (c) 480

45.One ticket is drawn at random from a bag containing tickets numbered 1 to 40. The probability that the selected ticket has a number which is a multiple of 5 is:

- (a) $1/5$ (b) $3/5$ (c) $4/5$ (d) $1/3$

Ans: (a) $1/5$

46Someone is asked to take a number from 1 to 100. The probability that it is a prime is:

- (a) $1/5$ (b) $6/25$ (c) $1/4$ (d) $13/50$

Ans: (c) $1/4$

47.A dice is thrown. Find the probability of getting an even number.

- (a) $2/3$ (b) 1 (c) $5/6$ (d) $1/2$

Ans: (d) $1/2$

48.Two coins are thrown at the same time. Find the probability of getting both heads.

- (a) $3/4$ (b) $1/4$ (c) $1/2$ (d) 0

Ans: (b) $1/4$

49.Two dice are thrown simultaneously. The probability of getting a sum of 9 is:

- (a) $1/10$ (b) $3/10$ (c) $1/9$ (d) $4/9$

Ans: (c) $1/9$

50. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, then the number of blue balls in a bag is:
(a) 5 (b) 10 (c) 15 (d) 20

Ans: (b) 10

51. Cards marked with numbers 2 to 101 are placed in a box and mixed thoroughly. One card is drawn from this box randomly, then the probability that the number on card is a perfect square.
(a) 9/100 (b) 1/10 (c) 3/10 (d) 19/100

Ans: (b) 1 / 100

52. What is the probability of getting 53 Mondays in a leap year?
(a) 1/7 (b) 53/366 (c) 2/7 (d) 7/366

Ans: (c) 2 / 7

A leap year consists of 366 days. There are 52 full weeks and 2 extra days.
There are 7 possible pairs for these extra 2 days:
(Monday, Tuesday) (Tuesday, Wednesday) (Wednesday, Thursday)
(Thursday, Friday) (Friday, Saturday) (Saturday, Sunday) (Sunday, Monday)

Using the probability formula : $P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}} \Rightarrow P = \frac{2}{7}$

53. A card is drawn from a well shuffled deck of 52 cards. Find the probability of getting a king of red suit.
(a) 1/26 (b) 3/26 (c) 7/52 (d) 1/13

Ans: (a) 1 / 26

54. A game of chance consists of spinning an arrow which is equally likely to come to rest pointing to one of the numbers 1,2,3.....12, then the probability that it will point to an odd number is:
(a) 1/6 (b) 1/12 (c) 7/12 (d) 5/12

Ans: (a) 1 / 6

55. Cumulative frequency curve is also called
(a) Histogram (b) Ogive (c) Bar graph (d) Median

Ans: (b) Ogive

56. The relationship between mean, median and mode for a moderately skewed distribution is

- (a) Mode = Median – 2 Mean (b) Mode = 3 Median – 2 Mean
(c) Mode = 2 Median – 3 Mean (d) Mode = Median – Mean

Ans: (b) Mode = 3 Median – 2 Mean

57. Mode and mean of a data are 12k and 15K. Median of the data is
(a) 12k (b) 14k (c) 15k (d) 16k

Ans: (b) 14 k

58. The abscissa of the point of intersection of the less than type and of the more than type cumulative frequency curves of a grouped data gives its
(a) Mean (b) Median (c) Mode (d) All the three above

Ans: (b) Median

59. The method used to find the mean of a given data is / are:
(a) Direct method (b) Assumed mean method
(c) Step deviation method (d) All the above

Ans: (d) All the above

60. While computing mean of grouped data, we assume that the frequencies are
(a) Evenly distributed over all the classes
(b) Centred at the class marks of the classes
(c) Centred at the upper limits of the classes
(d) Centred at the lower limits of the classes

Ans: (b) Centred at the class marks of the classes

61. What is Statistics?
(a) Statistics is the collection of data
(b) Statistics is the collection, classification and interpretation of data
(c) Statistics is the classification of data
(d) Statistics is the interpretation of data

Ans: (b) Statistics is the collection, classification and interpretation of data

62. What is Range?
(a) Largest value – Smallest value
(b) Smallest value – Largest value
(c) Mid value – Average value
(d) Average value – Mid value

Answer: (a) Largest value – Smallest value

63. While computing mean of grouped data, we assume that the frequencies are
(a) Centred at the upper limits of the classes
(b) Centred at the lower limits of the classes
(c) Centred at the class marks of the classes
(d) Evenly distributed over all the classes

Ans: (c) Centred at the class marks of the classes

64. If $x_1, x_2, x_3, \dots, x_n$ are the observations of a given data. Then the mean of the observations will be:
(a) Sum of observations / Total number of observations
(b) Total number of observations / Sum of observations
(c) Sum of observations + Total number of observations
(d) None of the above

Ans: (a) Sum of observations / Total number of observations

65. If the mean of frequency distribution is 7.5 and $\sum f_i x_i = 120 + 3k$, $\sum f_i = 30$, then k is equal to:
(a) 40 (b) 35 (c) 50 (d) 45

Ans: (b) 35

66. Which of the following cannot be determined graphically?
(a) Mean (b) Median (c) Mode (d) None of these

Ans: (a) Mean

67. Mode is the
(a) Middle most frequent value (b) Least frequent value
(c) Maximum frequent value (d) None of these

Ans: (c) Maximum frequent value

68. While computing mean of the grouped data, we assume that the frequencies are:
(a) Evenly distributed over all the classes
(b) Centred at the class marks of the classes
(c) Centred at the upper limits of the classes
(d) Centred at the lower limits of the classes

Ans: (b) Centred at the class marks of the classes

69. The abscissa of the point of intersection of the less than type and of the more than type cumulative frequency of a grouped data gives its:
(a) Mean (b) Median (c) Mode (d) All of these

Ans: (b) Median

70. If the arithmetic mean of x , $x + 3$, $x + 6$, $x + 9$ and $x + 12$ is 10, then $x = ?$
(a) 1 (b) 2 (c) 6 (d) 4

Ans: (d) 6

71. If the mean of first n natural numbers is $5n/9$, then $n = ?$
(a) 6 (b) 7 (c) 9 (d) 10

Ans: (c) 9

72. If 35 is removed from the data, 30, 34, 35, 36, 37, 38, 39, 40 then the median increases by:
(a) 2 (b) 1.5 (c) 1 (d) 0.5

Ans: (d) 0.5

73. The Median when it is given that mode and mean are 8 and 9 respectively, is:
(a) 8.57 (b) 8.67 (c) 8.97 (d) 9.24

Ans: (b) 8.67

74. The mean of the first 10 multiples of 6 is
(a) 3.3 (b) 33 (c) 34 (d) None of these

Ans: (b) 33

75. The median of set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set
(a) is increased by 2
(b) is decreased by 2
(c) is two times of the original number
(d) Remains the same as that of the original set.

Ans: (d) Remains the same as that of the original set.

76. Who gave the definition of probability?

- (a) Euclid (b) Simon Laplace (c) Archimedes (d) Einstein

Ans: (b) Simon Laplace

77. What is the sum of the probability of occurring an event E and the probability of not occurring the event E?

- (a) 0 (b) -1 (c) 1 (d) ∞

Ans: (c) 1

78. What is the name of the event for which the probability is zero?

- (a) Impossible event (b) Random event
(c) Exhaustive events (d) Mutual events

Ans: (a) Impossible event

79. What is the probability of a sure event?

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (b) 1

80. What kind of an event is getting a head or a tail when a coin is tossed?

- (a) Impossible event (b) Equal event
(c) Exhaustive event (d) Equally likely

Ans: (d) Equally likely

81. The probability of occurring an event is 0.45. Find the probability of not occurring the event.

- (a) 0.5 (b) 0.45 (c) 0.55 (d) 0.1

Ans: (c) 0.55

82. What is the probability of selecting 8 blue balls from 10 green balls?

- (a) 1 (b) 0 (c) 0.23 (d) 0.5

Ans: (b) 0

83. What is the median?

- (a) Difference between higher half and lower half of the data set
(b) Mean of the highest and lowest number in a data sample
(c) Value separating higher half from the lower half of a data sample
(d) Difference between the highest and lowest number.

Ans: (c) Value separating higher half from the lower half of a data sample

84. What is the Median of the following data sample? 2, 7, 4, 8, 9, 10, 6, 12, 13

- (a) 8 (b) 11 (c) 9 (d) 10

Ans: (a) 8

85. What is the Median of the following data sample? 3, 7, 4, 8, 9, 6, 10, 12, 13, 15

- (a) 7.5 (b) 9 (c) 8.5 (d) 1

Ans: (c) 8.5

86. Find the median of the given data samples. 90, 45, 67, 34, 26, 76, 44, 55.

- (a) 55 (b) 45 (c) 40 (d) 50

Ans: (d) 50

87.If the mean and the mode are given as 35 and 30. Find the Median.

- (a) 75 (b) 33.33 (c) 19 (d) 32

Ans: (b) 33.33

88.What is the formula for the median of Grouped data?

- (a) Median = $L + [(n / 2 - c f) / f] * h$
(b) Median = $L + [(n / 2 + c f) / f] * h$
(c) Median = $L + [(n / 2 - c f) / f] + h$
(d) Median = $L * [(n / 2 - c f) / f] * h$

Ans: (a) Median = $L + [(n / 2 - c f) / f] * h$

89.What is the mode in a data set?

- (a) The lowest number in a data set
(b) The number that occurs less frequently
(c) The number that occurs most frequently
(d) The highest number in a data set

Ans: (c) The number that occurs most frequently

90.What is the mode of 8, 5, 7, 10, 15, 21, 5, 7, 2, 5?

- (a) 2 (b) 5 (c) 7 (d) 21

Ans: (b) 5

91.Below are the results of the students in an exam. Find the mode of given results.

90, 80, 77, 86, 90, 91, 77, 25, 45, 35, 66, 69, 65, 43, 65, 75, 43, 90, 89.

- (a) 43 (b) 77 (c) 65 (d) 90

Ans: (d) 90

92.Find the mode of given samples. 47, 25, 89, 47, 89, 89, 47, 73, 29, 64, 95, 47.

- (a) 89 (b) 47 (c) 25 (d) No Mode

Ans: (b) 47

93.Which of the following data sets have the highest Mode?

Data set A: 11, 13, 14, 25, 30, 14, 14, 4.

Data set B: 22, 12, 27, 22, 17, 4, 7, 13.

- (a) Data set A (b) Data set B (c) Both data sets (d) No mode

Ans: (b) Data set B

94.Mean of 100 items is 49. It was discovered that three items which should have been 60, 70, 80 were wrongly read as 40, 20, 50 respectively. The correct mean is

- (a) 48 (b) 49 (c) 50 (d) 60

Ans: (c) 50

95.The Values of mean and median are given as 22.5 and 20 respectively. Find the approximate value of Mode.

- (a) 15 (b) 20 (c) 12 (d) 13.5

Ans: (a) 15

$\begin{aligned}\text{Mean} - \text{Mode} &= 3(\text{Mean} - \text{Median}) \\ 22.5 - \text{Mode} &= 3(22.5 - 20) \Rightarrow \text{Mode} = 15\end{aligned}$
--

96.If the Mean and Mode are 25, then find the Median.

- (a) 13 (b) 9 (c) 25 (d) 0

Ans: (c) 25

$$\begin{aligned}\text{Mean} - \text{Mode} &= 3(\text{Mean} - \text{Median}) \\ 25 - 25 &= 3(25 - \text{Median}) \Rightarrow \text{Median} = 25\end{aligned}$$

97.Find the value of mode if the mean and median are 25 and 20 respectively.

- (a) 12 (b) 18 (c) 10 (d) 17.5

Ans: (c) 15

$$\begin{aligned}\text{Mean} - \text{Mode} &= 3(\text{Mean} - \text{Median}) \\ 25 - \text{Mode} &= 3(25 - 20) \Rightarrow \text{Mode} = 10\end{aligned}$$

98.If the value of the mean is 10 and the median is 2 less than the mean. Find the Mode.

- (a) 2 (b) 3 (c) 4 (d) 5

Ans: (c) 4

$$\begin{aligned}\text{Mean} - \text{Mode} &= 3(\text{Mean} - \text{Median}) \\ 10 - \text{Mode} &= 3(10 - 8) \Rightarrow \text{Mode} = 4\end{aligned}$$

99.The probability of a non-leap year selected at random contain 53 Sunday is:

- (a) 5 / 28 (b) 1/7 (c) 2/7 (d) 53/365

Ans: (b) 1 / 7

Non-leap year: $365 = 52 \times 7 + 1$. There are 52 full weeks and **1 extra day**.

$$P(53 \text{ Sundays} | \text{Non-leap}) = 1/7$$

100.The probability that it will rain tomorrow is 0.85. What is the probability that it will not rain tomorrow

- (a) 0.25 (b) 0.145 (c) 3 / 20 (d) None of these

Ans: (c) 3 / 20

$$\begin{aligned}P(\text{not } A) &= 1 - P(A) \\ P(\text{not rain}) &= 1 - 0.85 = 0.15 = \frac{15}{100} = \frac{3}{20}\end{aligned}$$